

CHEM1211 Physical & Inorganic Chemistry

4-Atomic Structure

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MSE-22

Atom: An atom is the basic building block of all matter in the universe. It is the smallest possible unit of a chemical element.

Atomic Structure: Atomic structure refers to the specific arrangement of the tiny particles that make up an atom.

Democritus:

- Was the first person to come up with the idea of atom.
- Believed that all matter was composed of invisible particles called atom.
- He also believed that different atoms has different sizes
- have different properties.
- Others philosophers of that timer didn't agree with his theories.

John Dalton (1766-1844) ⇒

Dalton is the "Father of atomic theory".

- All matters is composed of extremely small particles called atoms.
- All atoms of a given element are identical having the same (I) Size (II) mass (III) chemical prop.
- Atoms of different element are different.

Dalton's Atomic theory

1. Matter is made of atoms. All matter is composed of extremely small, indivisible particles called atoms. Atoms cannt be created or destroyed.
2. Atoms of the same element are identical. All atoms in the given element are identical in mass, size and chemical properties. Atoms of the different elements differ in these properties.
3. Compounds are formed by combining atoms. When elements combined to form compounds, their atoms joined in fixed, simple whole-number ratios. $(1:1,2:3)$
4. Chemical reactions are rearrangement of atoms. In a chemical reaction, atoms are neither created nor destroyed. They are simply rearranged to form new substances.

Laws Dalton's Theory explained:

5. **Law of conservation of mass** → Atoms aren't created/destroyed in reactions.
6. **Law of Definite proportions** → Compounds has fixed atom ratios.
7. **Law of multiple proportions** → Elements can combine in different whole number ratios to form different compounds.

Limitation: Dalton's theory, through groundbreaking, had several flaws later corrected by modern science: ⇒ Atoms can be devided (into protons, neutrons, electrons) ⇒ Atoms of the same element have different masses ⇒ Atoms can be created/destroyed in nuclear

reactions ⇒ All atoms of an element always not identical. ⇒ He had no concept of the atom's internal structure.

Thomson Atomic model (1904): ⇒ Used cathod rays to prove that Daltons solid ball model could be broken into small particles. ⇒ Thomson is credited with discovering electrons. ⇒ Cathod rays had been used for some time before Thomsons experiment.

⇒ A cathod ray is a tube that has a piece of metal, called an electrode, at each end. Each electrode connected to a power source.

⇒ When the power is turned on, an electrode becomes charged and produces a stream of charged particles. They travel from the cathode accross the tube to the anode.

⇒ Thomson put the tube in a magnetic field. He predicted that the steam would travel in a straight path. Instead he found that the path curved away from a negatively charged plate. ⇒ Like charges repeal each other, unlike charges attract.

CATHODE RAYS – THE DISCOVERY OF ELECTRON

The knowledge about the electron was derived as a result of the study of the electric discharge in the **discharge tube** (J.J. Thomson, 1896). The discharge tube consists of a glass tube with metal electrodes fused in the walls (Fig. 1.1). Through a glass side-arm air can be drawn with a pump. The electrodes are connected to a source of high voltage (10,000 Volts) and the air partially evacuated. The electric discharge passes between the electrodes and the residual gas in the tube begins to glow. If virtually all the gas is evacuated from within the tube, the glow is replaced by faintly luminous 'rays' which produce fluorescence on the glass at the end far from the cathode. **The rays which proceed from the cathode and move away from it at right angles in straight lines are called Cathode Rays.**

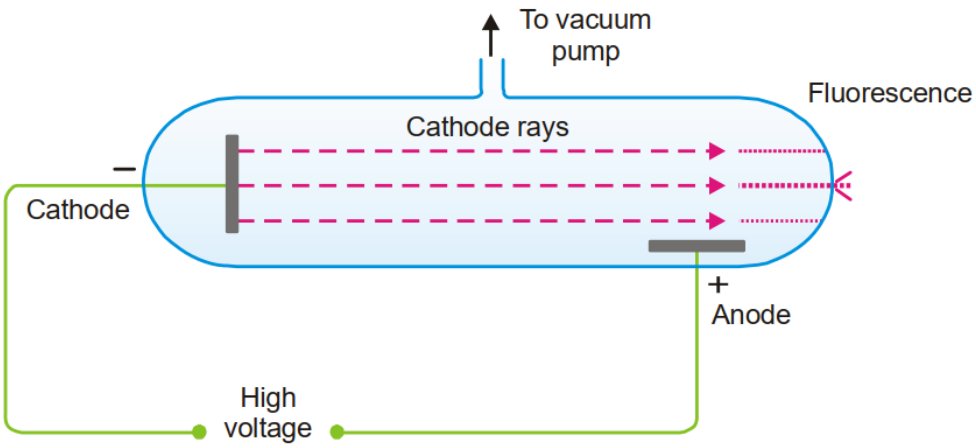


Figure 1.1
Production of cathode rays.

PROPERTIES OF CATHODE RAYS

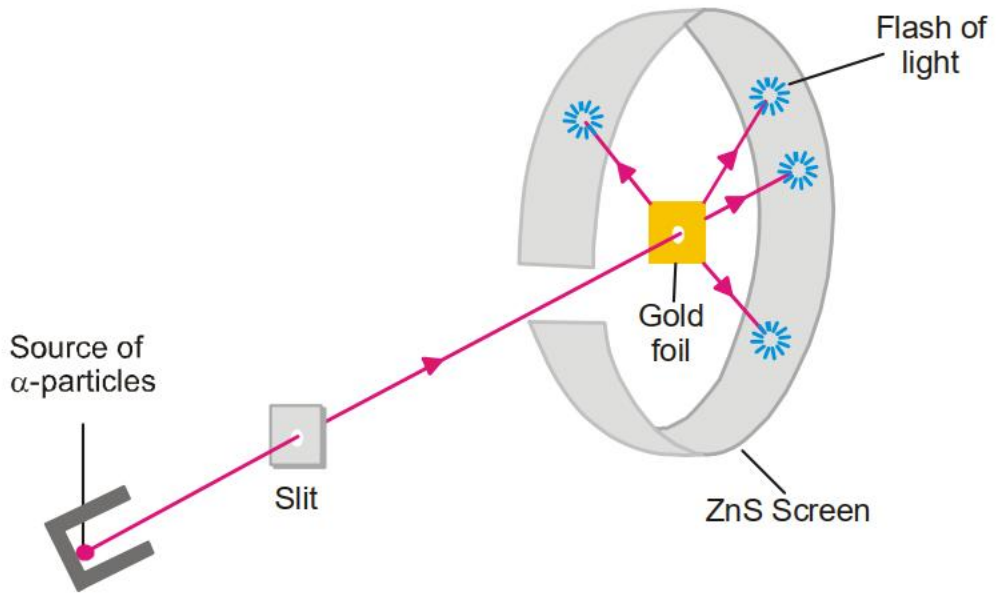
1. They travel in straight lines away from the cathode and cast shadows of metallic objects placed in their path.
2. Cathode rays cause mechanical motion of a small pin-wheel placed in their path. Thus they possess kinetic energy and must be material particles.
3. They produce fluorescence (a glow) when they strike the glass wall of the discharge tube.
4. They heat up a metal foil to incandescence which they impinge upon.
5. Cathode rays produce X-rays when they strike a metallic target.
6. Cathode rays are deflected by the electric as well as the magnetic field in a way indicating that they are streams of minute particles carrying negative charge.

Thomson concluded:

- Cathode rays are made up of invisible negatively charged particles called electrons.
- These electrons had to come from the matter (atoms) of the negative electrode.

RUTHERFORD'S ATOMIC MODEL – THE NUCLEAR ATOM

Having known that atom contains electrons and a positive ion, Rutherford proceeded to perform experiments to know as to how and where these were located in the atom. In 1909 Rutherford and Marsden performed their historic **Alpha Particle-Scattering Experiment,** using the apparatus illustrated in Fig. 1.6. They directed a stream of very highly energetic α-particles from a radioactive source against a thin *gold foil* provided with a circular fluorescent zinc sulphide screen around it. Whenever an α-particle struck the screen, a tiny flash of light was produced at that point



Rutherford and Marsden noticed that most of the α-particles passed straight through the gold foil and thus produced a flash on the screen behind it. This indicated that gold atoms had a structure with plenty of empty space. To their great astonishment, tiny flashes were also seen on other portions of the screen, some time in front of the gold foil. This showed that gold atoms deflected or 'scattered'

α-particles through large angles so much so that some of these bounced back to the source. Based on these observations, Rutherford proposed a model of the atom which is named after him. This is also

called the **Nuclear Atom.** According to it

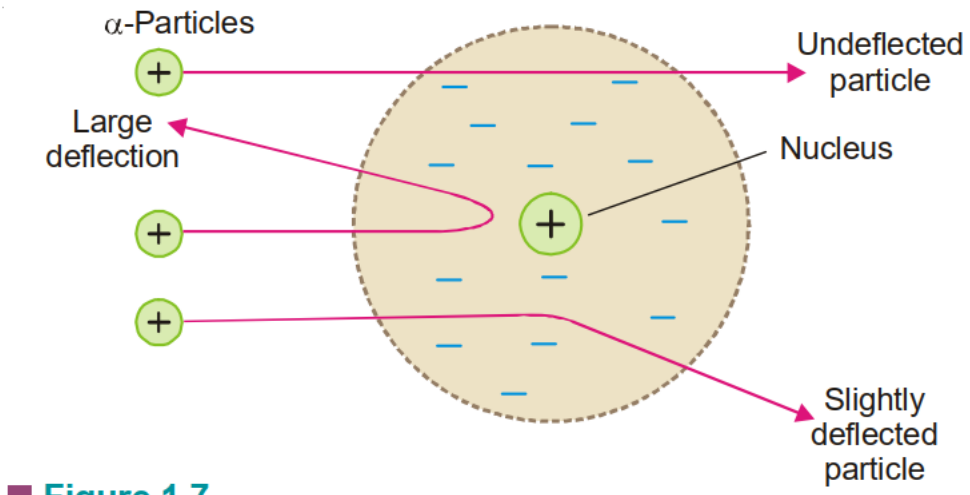


Figure 1.7
How nuclear atom causes scattering of α-particles.

(1) **Atom has a tiny dense central core or the nucleus which contains practically the entire mass of the atom, leaving the rest of the atom almost empty.** The diameter of the nucleus is about 10–13 cm as compared to that of the atom 10– 8 cm. If the nucleus were the size of a football, the entire atom would have a diameter of about 5 miles. It was this empty space around the nucleus which allowed the α-particles to pass through undeflected.

(2) **The entire positive charge of the atom is located on the nucleus, while electrons were distributed in vacant space around it.** It was due to the presence of the positive charge on the nucleus that α-particle (He2+) were repelled by it and scattered in all directions.

(3) **The electrons were moving in orbits or closed circular paths around the nucleus like planets around the sun.**

Postulates of Bohr Atomic Model

1. In an atom, electrons revolve around the nucleus in definite circular orbits.

2. As long as an electron is in its orbit, it neither absorbs nor radiates energy. Hence each orbit corresponds to a definite amount of energy. These orbits are called energy levels or stationary levels or energy shells denoted by an integer n, which have values of 1,2,3 … The values of $n = 1,2,3$ etc. are also represented as K, L, M …

3. The angular momentum of an electron (mvr) is quantized, i.e., the electron can revolve only in those orbits for which angular momentum of electron is an integral multiple of $h/2\pi$. Thus:

$$mvr = n \cdot \frac{h}{2\pi}$$

Where m is the mass of electron, v is the velocity of electron, r is the radius of orbit, h is Planck's constant, and n is an integer which is known as principal quantum number.

4. An electron can move from one energy level to another energy level by either absorbing or emitting radiation. Suppose an electron moves from higher energy level E_2 to lower energy level E_1, then the energy radiated is given by Planck's equation:

$$\Delta E = E_2 - E_1 = h\nu$$

$h =$ Planck's constant $\nu =$ frequency of radiation emitted

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Radius of Hydrogen-like Atom

For an electron of mass m, moving with a velocity v in an orbit of radius r:

$$\text{Centrifugal force} = \frac{mv^2}{r} \quad \text{— (1)}$$

$$\text{Coulombic attraction force} = \frac{Ze^2}{4\pi\epsilon_0 r^2} \quad \text{— (2)}$$

Here e is electronic charge, Z is the number of charges on nucleus, and ϵ_0 is the vacuum permittivity.

For an electron to remain in its orbit, the centrifugal force must be equal to the Coulombic attraction force. Hence:

$$\frac{mv^2}{r} = \frac{Ze^2}{4\pi\epsilon_0 r^2}$$

$$v^2 = \frac{Ze^2}{4\pi\epsilon_0 mr} \quad \text{— (3)}$$

According to one of the postulates of Bohr atomic model, the angular momentum of electron (mvr) is given by:

$$mvr = \frac{nh}{2\pi}$$

$$v^2 = \frac{n^2h^2}{4\pi^2m^2r^2} \quad \text{—(4)}$$

Combining equations (3) and (4):

$$\frac{Ze^2}{4\pi\epsilon_0 mr} = \frac{n^2h^2}{4\pi^2m^2r^2}$$

$$r = \frac{\epsilon_0 n^2h^2}{\pi mZe^2} \quad \text{—(5)}$$

For H-atom, $Z = 1$. Substituting the values of ϵ_0, h, m, and e:

$$r_n = n^2 \times 0.529 \times 10^{-10}\ \text{m} \quad \text{—(6)}$$

- When $n = 1$: $r_1 = 1^2 \times 0.529 \times 10^{-10}$ m
- When $n = 2$: $r_2 = 2^2 \times 0.529 \times 10^{-10}$ m
- When $n = 3$: $r_3 = 3^2 \times 0.529 \times 10^{-10}$ m

The radius of the 1st orbit of H-atom is often called the Bohr radius and expressed as r_0 or r_H, equal to 0.529×10^{-10} m.

For a hydrogen-like atom (Li^{2+}, He^{+}), the radius is:

$$r_n = \left(\frac{n^2}{Z}\right) a_H \quad \text{—(7)}$$

Energy of Electron in Hydrogen Atom

The energy of an electron in an orbit is given by:

$$E_n = \text{potential energy} + \text{kinetic energy}$$

$$E_n = -\frac{Ze^2}{4\pi\epsilon_0 r} + \frac{1}{2}mv^2$$

Replacing the value of mv^2 from Equation (3):

$$E_n = -\frac{Ze^2}{4\pi\epsilon_0 r} + \frac{1}{2} \cdot \frac{Ze^2}{4\pi\epsilon_0 r}$$

$$E_n = -\frac{1}{2} \cdot \frac{Ze^2}{4\pi\epsilon_0 r} = -\frac{Ze^2}{8\pi\epsilon_0 r} \quad \text{— (8)}$$

Putting the value of r in the above equation:

$$E_n = -\frac{Ze^2}{8\pi\epsilon_0} \times \frac{\pi m Ze^2}{\epsilon_0 n^2 h^2}$$

$$E_n = -\frac{mZ^2e^4}{8\epsilon_0^2 h^2} \cdot \frac{1}{n^2} \quad \text{— (9)}$$

When an electron moves from one energy level to another, the energy change is given by:

$$\Delta E = E_2 - E_1 = -\frac{mZ^2e^4}{8\epsilon_0^2 h^2}\left(\frac{1}{n_2^2} - \frac{1}{n_1^2}\right)$$

$$h\nu = \frac{mZ^2e^4}{8\epsilon_0^2 h^2}\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

$$\frac{hc}{\lambda} = \frac{mZ^2e^4}{8\epsilon_0^2 h^2}\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

$$\bar{\nu} = \frac{1}{\lambda} = \frac{mZ^2e^4}{8\epsilon_0^2 h^3 c}\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) = R_H \cdot \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) \quad \text{— (10)}$$

(Here, $Z = 1$ for H-atom)

R_H is called the Rydberg constant, which has a value of 1.097×10^7 m^{-1}.

From Equation (9):

$$E_n = -\frac{mZ^2e^4}{8\epsilon_0^2 h^2} \cdot \frac{1}{n^2} = -\frac{mZ^2e^4}{8\epsilon_0^2 h^3 c} \cdot \frac{hc}{n^2} = -\frac{R_H hc}{n^2} \quad \text{— (11)}$$

The above equation is used to calculate the energy of an electron in the n-th orbit.

Significance of Negative Value of Energy

The energy of an electron at infinity is arbitrarily assumed to be zero. This state is called *zero-energy* state. When an electron moves and comes under the influence of nucleus, it does some work and spends its energy in this process. Thus the energy of the electron decreases and it becomes less than zero *i.e.* it acquires a negative value.

Isotopes & Atomic Weight

- **Isotopes:** Isotopes are atoms having the same atomic number Z but different mass number.

- **Hydrogen:** $^{1}H, ^{2}H, ^{3}H$
- **Oxygen:** $^{16}O, ^{17}O, ^{18}O$
- **Carbon:** $^{12}C, ^{13}C, ^{14}C$
- **Uranium:** $^{235}U, ^{238}U$

Atomic weight of an element is a statistical mean of the atomic masses of different isotopes of that element available in nature.

Problem 1: The exact masses of ^{35}Cl and ^{37}Cl are 34.96885 and 36.96590 respectively, their abundances are 75.77% and 24.23% respectively. Calculate the atomic weight of chlorine atom.

- **ANS:**

$$\text{Atomic wt. of chlorine} = \frac{34.96885 \times 75.77 + 36.96590 \times 24.23}{100} = 35.45$$

Radioactivity

The phenomenon of spontaneous emission of particles (e.g., α- or β-particles) or γ-rays from a substance is called radioactivity. The substances which emit particles or rays are called radioactive substances. Examples: $^{238}_{92}U$, $^{232}_{90}Th$, $^{226}_{88}Ra$.

Disintegration Law

For a single radioactive decay process, the number of nuclei disintegrating in a short period of time is proportional to the number of nuclei present.

Let dN be the number of nuclei disintegrating in a small time dt, then:

$$\frac{dN}{dt} \propto -N \quad \text{or} \quad \frac{dN}{dt} = -\lambda N \quad \text{— (1)}$$

λ is the proportionality constant, known as the decay constant.

Rearranging and integrating equation (1):

$$\int \frac{dN}{N} = -\lambda \int dt \quad \text{or} \quad \ln N = -\lambda t + C \quad \text{— (2)}$$

At $t = 0, N = N_0$, thus $C = \ln N_0$.

From equation (2):

$$\ln N = -\lambda t + \ln N_0 \quad \text{or} \quad N = N_0 e^{-\lambda t} \quad \text{— (3)}$$

Where N_0 and N are the numbers of nuclei present at time zero and t respectively.

Half-Life Period

A quantity frequently used to characterize a radioelement is its half-life period ($t_{1/2}$). The half-life of a radioelement is the time needed to reduce the number of nuclei to half of its original number.

When $t = t_{1/2}, N = N_0/2$.

From equation (3):

$$\frac{N_0}{2} = N_0 e^{-\lambda t_{1/2}} \Longrightarrow \frac{1}{2} = e^{-\lambda t_{1/2}}$$

$$\lambda t_{1/2} = \ln 2 \Longrightarrow t_{1/2} = \frac{0.693}{\lambda} \quad -(4)$$

Thus, the half-life of a radioelement is inversely proportional to its decay constant.

Problems:

1. The half-life of tritium (^{3}H) is 12.3 years. If the initial amount of ^{3}H is 1.5 mg, how many milligrams remain after 49.2 years? *(ANS:* 0.094 mg*)*
2. A sample of radon initially emitted 7.0×10^4 α-particle disintegrations per second (dps). After 6.6 days, it emitted only 2.1×10^4 α-particles dps. What is the half-life of this isotope of radon? *(ANS:* 3.8 days*)*
3. The half-life of a radioactive substance is 200 years. What time is required to reduce the activity of 6.50×10^{12} disintegrations per minute (dpm) to an activity of 3.00×10^{-3} dpm? *(ANS:* 1.02×10^4 years*)*

Solutions of the Problems

1. Calculate the radius of 1st, 2nd and 3rd orbits of He^+ and Li^{2+} ions. Given that Bohr radius is 0.529×10^{-10} m.

Hint: Use the equation $r_n = \left(\frac{n^2}{Z}\right) a_H$.

1. He^+ $(Z = 2)$ **ion:**
   - For 1st orbit $(n = 1)$:

$$r_1 = \frac{1^2}{2} \times 0.529 \times 10^{-10} = 2.645 \times 10^{-11}\ \text{m}$$

   - For 2nd orbit $(n = 2)$:

$$r_2 = \frac{2^2}{2} \times 0.529 \times 10^{-10} = 1.058 \times 10^{-10}\ \text{m}$$

   - For 3rd orbit $(n = 3)$:

$$r_3 = \frac{3^2}{2} \times 0.529 \times 10^{-10} = 2.38 \times 10^{-10}\ \text{m}$$

   Li^{2+} $(Z = 3)$ **ion:**

- For 1st orbit ($n = 1$):

$$r_1 = \frac{1^2}{3} \times 0.529 \times 10^{-10} = 1.76 \times 10^{-11}\ \text{m}$$

- For 2nd orbit ($n = 2$):

$$r_2 = \frac{2^2}{3} \times 0.529 \times 10^{-10} = 7.05 \times 10^{-11}\ \text{m}$$

- For 3rd orbit ($n = 3$):

$$r_3 = \frac{3^2}{3} \times 0.529 \times 10^{-10} = 1.59 \times 10^{-10}\ \text{m}$$

2. **Energy calculations:**

2. What are the energies of $n = 1$ and $n = 2$ states of H-atom in (a) J/atom and (b) kJ mol^{-1}? Given that $R_H = 1.097 \times 10^7\ \text{m}^{-1}$.

Hint: Use the equation $E_n = -\frac{R_H hc}{n^2}\ \text{J/atom}$ and then convert into kJ mol^{-1}.

- **(a)**

$$E_1 = -\frac{R_H hc}{n^2}\ \text{J/atom}$$

$$E_1 = -\frac{1.097 \times 10^7\ \text{m}^{-1} \times 6.626 \times 10^{-34}\ \text{J} \cdot \text{s} \times 2.9979 \times 10^8\ \text{m} \cdot \text{s}^{-1}}{1^2}$$
$$= -2.18 \times 10^{-18}\ \text{J/atom}$$

$$E_2 = -\frac{E_1}{2^2} = -5.45 \times 10^{-19}\ \text{J/atom}$$

- **(b)**

$$E_1 = -\left(\frac{2.18 \times 10^{-18}\ \text{J}}{\text{atom}}\right)\left(\frac{6.023 \times 10^{23}\ \text{atoms}}{\text{mol}}\right)\left(\frac{1\ \text{kJ}}{1000\ \text{J}}\right) = -1312\ \text{kJ mol}^{-1}$$

$$E_2 = -328\ \text{kJ mol}^{-1}$$

3. **Wavelength (λ) & Wave Number ($\bar{\nu}$):**

3. Calculate the frequencies in wave numbers per meter and the wavelengths in nm of light emitted from H-atom when the electron undergoes a transition ($R_H = 1.097 \times 10^7 \text{m}^{-1}$): (i) from energy level $n = 3$ to level $n = 1$; (ii) from energy level $n = 3$ to level $n = 2$; (iii) from energy level $n = 4$ to level $n = 2$.

Hint: Use the equation $\bar{\nu} = \frac{1}{\lambda} = R_H\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$.

- **(i)**

$$\bar{\nu} = R_H\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) = 1.097 \times 10^7\left(\frac{1}{1^2} - \frac{1}{3^2}\right)\ \text{m}^{-1} = 9.75 \times 10^6\ \text{m}^{-1}$$

$$\lambda = \frac{1}{\bar{\nu}} = \frac{1}{9.75 \times 10^6} = 102.5 \times 10^{-9}\ \text{m} = 102.5\ \text{nm}$$

- **(ii)**

$$\bar{\nu} = 1.097 \times 10^7\left(\frac{1}{2^2} - \frac{1}{3^2}\right) = 1.52 \times 10^6\ \text{m}^{-1}$$

$$\lambda = \frac{1}{1.52 \times 10^6} = 656.3 \times 10^{-9}\ \text{m} = 656.3\ \text{nm}$$

- **(iii)**

$$\bar{\nu} = 1.097 \times 10^7 \left(\frac{1}{2^2} - \frac{1}{4^2}\right) = 2.06 \times 10^6\ \text{m}^{-1}$$

$$\lambda = 486.1 \times 10^{-9}\ \text{m} = 486.1\ \text{nm}$$

$$\Delta E = h\nu = \frac{hc}{\lambda} = \frac{6.626 \times 10^{-34}\ \text{J} \cdot \text{s} \times 2.9979 \times 10^8\ \text{m} \cdot \text{s}^{-1}}{589 \times 10^{-9}\ \text{m}} = 3.37 \times 10^{-19}\ \text{J}$$

$$\bar{\nu} = R_H \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

- For 1st line of Lyman series ($n_1 = 1, n_2 = 2$):

$$\bar{\nu} = 1.097 \times 10^7 \left(\frac{1}{1^2} - \frac{1}{2^2}\right) = 8.23 \times 10^6\ \text{m}^{-1}$$

- For 2nd line ($n_1 = 1, n_2 = 3$):

$$\bar{\nu} = 1.097 \times 10^7 \left(\frac{1}{1^2} - \frac{1}{3^2}\right) = 9.75 \times 10^6\ \text{m}^{-1}$$

4. Calculate the frequencies of 1st and 2nd spectral line of Balmer series.

- For 1st line of Balmer Series ($n_1 = 2, n_2 = 3$):

$$\bar{\nu} = 1.097 \times 10^7 \left(\frac{1}{2^2} - \frac{1}{3^2}\right) = 1.52 \times 10^6\ \text{m}^{-1}$$

- For 2nd line ($n_1 = 2, n_2 = 4$):

$$\bar{\nu} = 1.097 \times 10^7 \left(\frac{1}{2^2} - \frac{1}{4^2}\right) = 2.06 \times 10^6\ \text{m}^{-1}$$

$$\lambda = \frac{h}{mv} = \frac{6.626 \times 10^{-34}\ \text{J} \cdot \text{s}}{9.109 \times 10^{-31}\ \text{kg} \times 2.5 \times 10^7\ \text{m} \cdot \text{s}^{-1}} = 2.91 \times 10^{-11}\ \text{m}$$

Origin of Hydrogen Spectra

When H-atoms are excited, many of their electrons are raised to higher energy levels. When such electrons come back to lower energy levels, a series of spectral lines is observed.

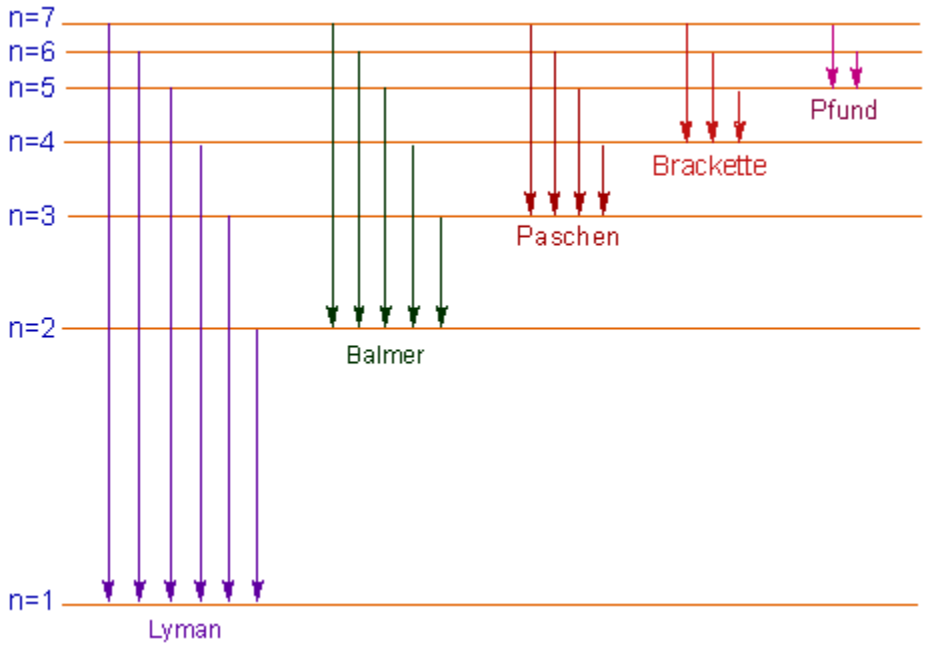
- When electrons come back to the 1st energy level from higher energy levels, the **Lyman series** appears in the ultraviolet region.
- The **Balmer series** appears in the visible region when electrons come back to the 2nd energy level.

- **Paschen**, **Brackett**, and **Pfund** series are observed in the infrared region when electrons drop back into the 3rd, 4th, and 5th energy levels respectively.

The frequencies of different spectral lines of all series can be calculated by:

$$\bar{\nu} = \frac{1}{\lambda} = R_H \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

Series	n_1	n_2	Region of Spectrum
Lyman	1	2,3,4 ...	Ultraviolet
Balmer	2	3,4,5 ...	Visible
Paschen	3	4,5,6 ...	Infrared
Brackett	4	5,6,7 ...	Infrared
Pfund	5	6,7,8 ...	Infrared



Limitations of Bohr Model

1. Bohr model cannot explain how radiation occurs when an electron jumps from one energy level to another energy level.
2. The Bohr model does not explain fine structure in spectral lines.
3. The model successfully explains the spectra for hydrogen and hydrogen-like systems, but cannot explain the spectra of poly-electronic systems.
4. The model does not imply the wave nature of electrons.

5. It does not explain the Heisenberg Uncertainty Principle and the Zeeman Effect.

Arnold Sommerfeld proposed two major modifications to Bohr's atomic model in 1916 to address its limitations—specifically, its inability to explain the fine structure of hydrogen spectral lines and the spectra of complex atoms.

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1. Elliptical Orbits

Bohr assumed that electrons revolve around the nucleus only in **circular orbits**. Sommerfeld proposed that circular orbits are merely a special case of a broader class of **elliptical orbits**.

When an electron revolves in an elliptical orbit:

- The nucleus resides at one of the foci of the ellipse.
- The distance between the electron and the nucleus (r) and the azimuthal angle (ϕ) vary continuously.

To define these elliptical orbits, Sommerfeld introduced two quantum numbers instead of Bohr's single principal quantum number (n):

- **Principal Quantum Number (n):** Determines the main energy level and size of the orbit.
- **Azimuthal (or Orbital) Quantum Number (k or l):** Determines the shape (ellipticity) of the orbit.

The ratio of the semi-major axis (a) to the semi-minor axis (b) of the elliptical path is governed by the relation:

$$\frac{a}{b} = \frac{n}{k}$$

Where k takes integer values from 1 to n:

- When $k = n$, $a = b$, meaning the orbit is **circular**.
- When $k < n$, $a > b$, making the orbit **elliptical**. The smaller the value of k, the more eccentric (flattened) the ellipse.

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2. Relativistic Mass Variation

Bohr treated the electron's speed and mass as constant. Sommerfeld applied Einstein's Special Theory of Relativity to account for the fact that an electron moves at extremely high speeds, especially when it passes close to the heavy nucleus in highly elliptical orbits.

According to relativity, an object's mass changes with velocity:

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Where:

- m_0 is the rest mass of the electron.
- v is the velocity of the electron.
- c is the speed of light.

Because the velocity v varies continuously along an elliptical trajectory, the mass m changes along the path. As a consequence, the elliptical orbit is not closed; it precesses around the nucleus, forming a **rosette-like path**.

Main Achievement & Success

- **Explanation of Fine Structure:** Due to relativistic mass variation, energy levels associated with the same principal quantum number n split into slightly different sub-levels based on different k values. Transitions between these closely spaced sub-levels explained the fine splitting (fine structure) observed in the hydrogen spectrum.

Limitations of Sommerfeld's Model

- It could not predict the correct number of observed spectral lines in Zeeman and Stark effects without further quantum mechanics.
- It failed to explain the spectra of multi-electron atoms.
- It still relied on classical trajectories, violating the **Heisenberg Uncertainty Principle** that was developed later.

De Broglie Equation: Dual Nature of Particle

The energy of a photon is given by Planck's equation:

$$E = h\nu \quad \text{— (1)}$$

Einstein's equation relating matter and energy:

$$E = mc^2 \quad \text{— (2)}$$

Equating the two equations:

$$h\nu = mc^2 \Longrightarrow h\frac{c}{\lambda} = mc^2 \Longrightarrow \frac{h}{\lambda} = mc \Longrightarrow \lambda = \frac{h}{mc} \quad -(3)$$

Replacing the speed of light c with v (velocity of a particle of mass m) for real particles gives the **de Broglie equation**:

$$\lambda = \frac{h}{mv} \quad -(4)$$

De Broglie Relationship and Bohr's Theory

For a wave to remain continuously in phase, the circumference of the orbit ($2\pi r$) must be an integral multiple of wavelength (λ):

$$n\lambda = 2\pi r \Longrightarrow n\left(\frac{h}{mv}\right) = 2\pi r \Longrightarrow mvr = n\frac{h}{2\pi}$$

Thus, de Broglie's equation supports Bohr's postulate of the quantization of electron angular momentum.

Periodic Table and Periodic Properties

Mendeleev's Periodic Law: The physical and chemical properties of the elements are a periodic function of their atomic weight.

Defects of Mendeleev's periodic table:

5. The lanthanides and actinides have not been placed properly in the periodic table.
6. The elements of higher atomic weight precede those of lower atomic weight in the following three pairs of elements:
    - Ar (39.95) before K (39.09)
    - Co (58.93) before Ni (58.69)
    - Te (127.6) before I (126.9)
7. It is impossible to accommodate large number of isotopes in the periodic table.

Most of the defects of Mendeleev's periodic table disappear if the elements are arranged on the basis of atomic number.

Modern periodic law: The properties of the elements are a periodic function of their atomic number, i.e., the properties of the elements are repeated after definite regular intervals.

The Classification of Elements on the Basis of Electronic Configuration

1. Differentiating Electron Classification

The elements are divided into s, p, d, and f-blocks depending on the nature of the atomic orbitals into which the last electron (called differentiating electron) enters.

- **s-block elements:** In these elements the last electron enters into the ns orbitals. The elements of IA (alkali metals) and IIA (alkaline earth metals) groups belong to this block. The valence shell electronic configuration of these elements varies from ns^1 (IA group) to ns^2 (IIA group). These elements are strongly electropositive metals. They lie on the extreme left of the periodic table.
- **p-block elements:** The elements in which p-orbitals are being progressively filled are called p-block elements. The elements from group IIIA to VIIA and zero (except helium) are included in this block. The valence shell electronic configuration of these elements is ns^2np^{1-6} (except helium whose configuration is $1s^2$). These elements lie on the extreme right of the periodic table and consist of some metals, all non-metals, metalloids, and noble gases.
- **d-block elements:** The elements where the last electron goes into d-orbitals of penultimate shell are called d-block elements or transition elements. The valence shell configuration of these elements can be represented as $(n-1)d^{1-10}ns^{1-2}$. They are classified into four series viz. 3d, 4d, 5d, and 6d corresponding to the filling of 3d, 4d, 5d, and 6d orbitals respectively. All are metals and have high melting and boiling points. These are good conductors of heat and energy. The elements are located in the middle of the periodic table, i.e., between s and p block elements.
- **f-block elements:** The elements where f-orbitals are being filled are called f-block elements. The valence shell configuration of these elements can be represented as $(n-2)f^{0-14}(n-1)d^{0-2}ns^2$. These elements are also called inner transition elements and are classified into two series viz. 4f series (lanthanides) and 5f series (actinides). The elements are located in group IIIB and placed at the bottom of the periodic table.

2. Bohr's Classification

In accordance to Bohr the elements have been grouped into four types:

- **Inert gases:** In the atoms of these elements the outermost s and p subshells are completely filled. Thus the valence shell configuration of these elements is ns^2np^6 with the exception of helium which has $1s^2$ configuration. Due to highly stable configuration these elements do not show chemical activity under normal condition and thus termed as inert gases. He, Ne, Ar, Kr, Xe, and Rn belong to this group.
- **Representative elements:** In the atoms of these elements the outermost shell is incomplete. The valence shell configuration of these elements varies from ns^1 to ns^2np^5. These consist of some metals, all non-metals and metalloids. These elements are found in nature in abundance and active in nature. On this account,

these elements are called representative elements. Number of electrons present in outermost shell shows the group to which these elements belong.

- **Transition elements (d-block elements):** In the atoms of these elements the outermost shell and the penultimate shell (next to the outermost) are incomplete. The valence shell configuration of these elements can be represented as $(n-1)d^{1-10}ns^{0,1,2}$. They are classified into four series viz. 3d, 4d, 5d, and 6d.
- **Inner transition elements (f-block elements):** Atoms of these elements have three outermost shells incomplete. The valence shell configuration of these elements can be represented as $(n-2)f^{0-14}(n-1)d^{0-2}ns^2$. The elements are classified into two series viz. 4f series (lanthanides) and 5f series (actinides).

Periodic Properties

Atomic or Ionic Radii

The atomic or ionic radius is the distance between the nucleus and the outermost electron shell of the atomic or ionic species. Since it is impossible to isolate an individual atom, it is not possible to measure the atomic radius directly. Hence, covalent radius is preferable to atomic radius.

Covalent Radius

It is defined as half the distance between the nuclei of two like atoms bonded together by a single covalent bond. For a homonuclear diatomic molecule of A_2 type, the covalent radius is measured by dividing the A-A bond length by two. The Cl-Cl bond length in Cl_2 molecule is 198 pm. Thus, the covalent radius of Cl is $198/2 = 99$ pm.

For a heteronuclear diatomic molecule of AB type, the covalent radius is measured by an indirect method. The C-Cl bond length in CCl_4 molecule is 176 pm. Thus:

$$d(\text{C-Cl}) = r(\text{C}) + r(\text{Cl})$$

$$176\ \text{pm} = r(\text{C}) + 99\ \text{pm}$$

$$r(\text{C}) = (176 - 99)\ \text{pm} = 77\ \text{pm}$$

Periodic Variation of Atomic / Ionic Radii

Atomic and ionic radii both decrease from left to right in a period, and on moving down a group, covalent and ionic radii increase.

In a period, with the addition of an electron in the same energy level, the nuclear charge also increases by one. The increased nuclear charge attracts the electrons more strongly towards the nucleus, and thus the size of atoms decreases.

Elements of 2nd Period:

Element	Be	B	C	N	O	F
Covalent radius (pm)	112	90	77	75	73	71

On moving down a group, the main energy level increases and the outermost electron shells get farther away from the nucleus. Thus the electrostatic attraction between the outermost electron shell and the nucleus decreases, which causes the increase of covalent radii.

Elements of Group IIA:

Element	Be	Mg	Ca	Sr	Ba
Covalent radius (pm)	112	160	197	215	224
Ionic radius, M^{2+} (pm)	31	72	100	113	136

The size of a cation is always smaller than its parent element. For example, Li^+ ion (60 pm) is smaller than the Li atom (152 pm). The size of an anion is always larger than its parent element. For example, Cl^- ion (181 pm) is larger than the Cl atom (99 pm).

Isoelectronic Species

Atoms or ions with the same number of electrons are called isoelectronic species. For example, O^{2-}, F^-, Na^+ and Mg^{2+} are isoelectronic. For such species, the size decreases with the increase of atomic number.

Ion	O^{2-}	F^-	Na^+	Mg^{2+}
Ionic radius (pm)	140	133	102	72
No. of protons in nucleus	8	9	11	12
No. of electrons	10	10	10	10

Ionization Energy (IE)

The ionization energy is defined as the energy required to remove the most loosely bound electron (i.e. the outermost electron) from an isolated gaseous atom of an element in its ground state.

$$X(g) + \text{energy supplied (IE)} \rightarrow X^+(g) + e^-$$

Periodic Variation

For main-group elements, the first IE generally increases across a period and decreases down a periodic group.

Elements of 2nd Period:

Element	Li	Be	B	C	N	O	F	Ne
IE (kJ mol^{-1})	520	897	801	1086	1402	1314	1681	2081

In a period, due to the increase of effective nuclear charge, the electrostatic attraction between the outermost electrons and the nucleus increases. Therefore, it is comparatively more difficult to remove an electron, and consequently the IE increases along a period.

On moving down a group, the size of atoms increases and the outermost electron shells go farther away from the nucleus. Therefore, less energy is needed to remove the outermost electron, and consequently the IE decreases.

Elements of IIA Group:

Element	Be	Mg	Ca	Sr	Ba
IE (kJ mol^{-1})	897	738	590	550	503

Explanation of Exceptions

The IE of B ($Z = 5, 1s^2 2s^2 2p^1$) is lower than that of Be ($Z = 4, 1s^2 2s^2$). This is because the $2p$ orbital of B is higher in energy than the $2s$ orbital of Be, so less energy is needed to remove the $2p$ electron of B than the $2s$ electron of Be. Moreover, the outermost electron in Be is paired, while the outermost electron in B is unpaired; since it is easier to remove an unpaired electron than a paired one, the IE of Be is higher than that of B. For the same reason, the IE of Mg ($Z = 12$, $[\text{Ne}]3s^2$) is higher than that of Al ($Z = 13$, $[\text{Ne}]3s^2 3p^1$).

The IE of N ($Z = 7, 1s^2 2s^2 2p^3$) is higher than that of O ($Z = 8, 1s^2 2s^2 2p^4$). According to Hund's rule, half-filled or completely filled orbitals are more stable. Since N has a half-filled electronic configuration, more energy is needed to remove an electron from the N atom than the O atom, so N has a higher IE value than O. Electron–electron repulsion also causes the O atom to have a lower IE value than expected. The IE of P ($Z = 15$, $[\text{Ne}]3s^2 3p^3$) is higher than that of S ($Z = 16$, $[\text{Ne}]3s^2 3p^4$) for the same reason.

Successive Ionization Energies

The energy required to remove the second electron from a cation is called the second IE:

$$X(g) + IE_1 \rightarrow X^+(g) + e^-$$

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$$X^+(g) + IE_2 \rightarrow X^{2+}(g) + e^-$$

$$X^{2+}(g) + IE_3 \rightarrow X^{3+}(g) + e^-$$

$$IE_1 < IE_2 < IE_3 \ldots$$

The successive increase in the values of IE is due to the fact that it is relatively more difficult to remove an electron from a cation of higher positive charge than a cation of lower positive charge, or a neutral atom.

Electron Affinity (EA)

The electron affinity (EA) is defined as the amount of energy released when an electron is added to an isolated gaseous atom to produce an anion.

$$A(g) + e^- \rightarrow A^-(g) + \text{energy released (EA)}$$

Since energy is released, the process is exothermic. The energy needed to convert $A^-(g)$ to $A^{2-}(g)$ is called the second electron affinity of A. In this process, an electron is added against the electrostatic repulsion between the incoming electron and the anion (A^-). Therefore, instead of energy being released, energy must be supplied, and the second electron affinity is endothermic.

Periodic Variation

The values of electron affinity increase from left to right in a period. On moving from left to right in a period, the atomic size decreases; thus the attraction of the nucleus towards the outside electron increases, hence electron affinity increases.

2nd Period:

Element	B	C	N	O	F
EA (kJ mol^{-1})	-27	-122	-7	-141	-328

On moving down a group, the atomic size increases steadily. Thus the attraction of the nucleus towards the outside electron decreases, hence the electron affinity decreases.

VIIA Group:

Element	F	Cl	Br	I	At
EA (kJ mol^{-1})	-328	-349	-324	-295	-270

Exceptions

According to the general rule, the electron affinity value of the F atom should be higher than that of the Cl atom. Actually, the electron affinity of F is lower than that of Cl. This is because the atomic size of F is smaller than that of Cl: the addition of an extra electron

produces electron-electron repulsion in the relatively compact $2p$ orbital of F. Due to this increasing repulsion, the F atom has a lesser tendency to attract the outside electron and consequently has a lower EA value. This exception is observed among the pairs of elements of the second and third period:

2nd Period:

Element	B	C	N	O	F
EA (kJ mol^{-1}**)**	-27	-122	-7	-141	-328

3rd Period:

Element	Al	Si	P	S	Cl
EA (kJ mol^{-1}**)**	-43	-134	-72	-200	-349

Electronegativity

The electronegativity of a bonded atom is defined as the relative tendency to attract the shared electron pair towards itself.

Periodic Variation

Electronegativity increases from left to right in a period. This is due to the increase of nuclear charge and decrease of atomic size; smaller atoms have a greater tendency to attract electrons towards themselves, i.e., smaller atoms have higher values of electronegativity.

2nd Period:

Element	Li	Be	B	C	N	O	F
Covalent radius (pm)	152	112	90	77	75	73	71
Electronegativity	1.0	1.5	2.0	2.5	3.0	3.5	4.0

Electronegativity values decrease from top to bottom in a group. As we move down a group, there is an increase of atomic size; with the increase of atomic size, electronegativity values decrease.

VIIA Group:

Element	F	Cl	Br	I	At
Electronegativity	4.0	3.0	2.8	2.5	2.2

The periodic variation of electronegativity is applicable for s- and p-block elements.

THE ENERGIES OF ORBITALS (ORDER OF FILLING ORBITALS WITH ELECTRONS)

The relative energy of sublevels will now determine the arrangement of electrons in atoms (electrons moving into the apartment block).

The **ground state** of an atom is the state in which the electrons are located in the lowest energy levels and sublevels possible. They can move to higher levels or an **excited state** if excited by some form of energy (light). Sublevels of energy levels overlap. A $4s$ subshell has lower energy than a $3d$ subshell. The effect of the overlap is that the atom is more stable when the $4s$ sublevel is of lower energy than the $3d$ sublevel. When drawing orbital diagrams (placing electrons in orbitals of increasing energy), electrons occupy sublevels of the lowest energy.

In drawing orbital filling diagrams, the following guidelines must be followed:

8. **THE AUFBAU PRINCIPLE (Building up principle):** ELECTRONS ENTER ORBITALS OF LOWEST ENERGY FIRST. Electrons are added to atomic orbitals starting with the lowest energy orbital and building up to higher energy orbitals.
9. **PAULI EXCLUSION PRINCIPLE:** ONLY 2 ELECTRONS CAN OCCUPY AN ORBITAL AND THEY MUST HAVE OPPOSITE SPINS. No two electrons in an atom can have the same four quantum numbers. This means that electrons occupying the same orbital cannot have the same spin (no 2 electrons can be in the same place at the same time).
    - **THIS:** [↑↓]
    - **NOT THIS:** [↑↑]

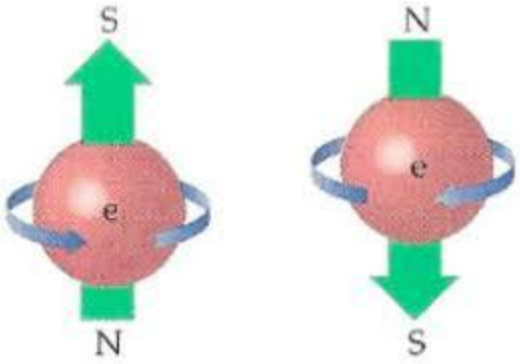


Figure : A picture of the spinning electron. Spinning in one direction, the electron produces the magnetic field oriented as shown in (a). Spinning in the opposite direction, it gives a magnetic field of the opposite orientation, as shown in (b).

3. **HUND'S RULE:** WHEN ELECTRONS OCCUPY ORBITALS OF EQUAL ENERGY, ONE ELECTRON ENTERS EACH ORBITAL UNTIL ALL THE ORBITALS CONTAIN ONE ELECTRON WITH SPINS PARALLEL. The most stable arrangement of electrons in subshells is the one with the greatest number of parallel spins, without violating the

Pauli Exclusion Principle. (Electrons occupy orbitals in a sublevel one at a time until each orbital has 1 electron. They will not double up until each orbital has at least one electron.)

- **THIS:** [↑][↑][↑]
- **NOT THIS:** [↑↓][↑][ ]
- **NOT THIS:** [↑][↓][↑]

****NOTE:** Orbitals in the same subshell are of equal energy, so an electron may occupy any of the orbitals in that subshell with equal probability. This means that the first orbital on the left is not necessarily the first one filled. By convention, we use this as the first orbital filled. Also, the first electron in a sublevel may be spin up [↑] or spin down [↓] with equal probability. We designate the first electron to be spin up [↑] by convention. Once the first one in has a spin [↑] or [↓], the remaining electrons in an orbital must have the same spin according to Hund's Rule.

- **All these are of equal probability:** [↑][↑][↑], [ ][↑][↑], etc.
- **These are not correct:** [↑][↓][↑], [↑↓][↑][ ].

ORDER OF FILLING SUBLEVELS

The **Diagonal Rule** can be used to determine which order the sublevels fill (shown in diagram). A better way to do it is to use the periodic table.

In using the periodic table:

- Period numbers are the level number.
- Go from left to right. Designate each period and sublevel ex. $2p, 3s$.
- Sublevel d is one level lower than the main energy level $(n-1)$.
- Sublevel f is two levels lower than the main energy level $(n-2)$.

ORDER IS: $1s \rightarrow 2s \rightarrow 2p \rightarrow 3s \rightarrow 3p \rightarrow 4s \rightarrow 3d \rightarrow 4p \rightarrow 5s \rightarrow 4d \rightarrow 5p \rightarrow 6s \rightarrow 4f \rightarrow 5d \rightarrow 6p \rightarrow 7s \rightarrow 5f \rightarrow 6d \rightarrow 7p$

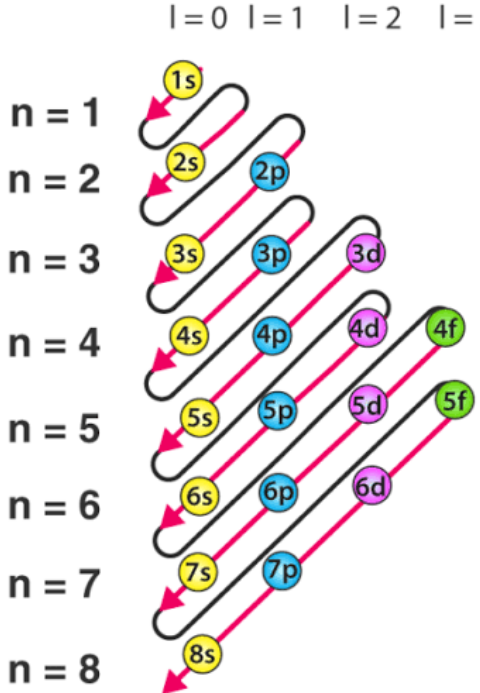


FIGURE : The order in which atomic subshells are filled in a many-electron atom. Start with the $1s$ orbital and move downward, following the direction of the arrows. Thus the order goes as follows: $1s < 2s < 2p < 3s < 3p < 4s < 3d < \cdots$ The direction of the arrow indicates the order in which atomic orbitals are filled according to the Aufbau principle.

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QUANTUM NUMBERS

Quantum numbers are the identification numbers that completely describe the position and energy of an individual electron in an atom.

1 **Principal Quantum Number (n):** This quantum number determines the main energy level or energy shell to which an electron belongs. It can have any positive integer values $1,2,3,$ …. The energy of an electron in an atom depends mainly on n. The smaller n is, the lower the energy. The size of an energy shell also depends on n. The larger the value of n, the greater the average distance of the shell from the center of the nucleus. Shells are sometimes denoted by letters:

- $n = 1 \rightarrow$ K
- $n = 2 \rightarrow$ L
- $n = 3 \rightarrow$ M
- $n = 4 \rightarrow$ N

The total number of orbitals for a given value of n is n^2. The maximum number of electrons possible in a given shell is $2n^2$.

2 **Angular Momentum (Secondary or Azimuthal) Quantum Number (l):** This quantum number divides the shells into subshells where the electron can be found. It gives the idea of the shape of the orbitals. Within a particular shell of quantum number n, there are different kinds of orbitals. The allowed values of l are integers from 0 to $n - 1$. For example, for $n = 3$, the possible values of l are $0,1,$ and 2. The different subshells are denoted by letters:

- $l = 0 \rightarrow s$
- $l = 1 \rightarrow p$
- $l = 2 \rightarrow d$
- $l = 3 \rightarrow f$
- $l = 4 \rightarrow g$
- $l = 5 \rightarrow h$ …

The subshell with $n = 2$ and $l = 1$ is the $2p$ subshell; if $n = 3$ and $l = 0$, it is the $3s$ subshell, and so on. Maximum number of electrons in a subshell (s, p, d, f, etc.) is equal to $2(2l + 1)$.

3 **Magnetic Quantum Number (m_l):** This number specifies the orientation in space of an orbital within the subshell. It also specifies to which orbital within a subshell the electron is located. The values of m_l can range from $-l$ to $+l$ including 0. There are $2l + 1$ orbitals in each subshell. Thus:

- s subshell ($l = 0, m_l = 0$) has 1 orbital.
- p subshell ($l = 1, m_l = -1,0,+1$) has 3 orbitals.
- d subshell ($l = 2, m_l = -2,-1,0,+1,+2$) has 5 orbitals, etc.

Orbitals of a particular subshell have the same energy and these orbitals are called **degenerate orbitals**.

4 **Spin Quantum Number (m_s):** This quantum number specifies the orientation of the spin axis of an electron. An electron can spin in only one of two directions. Thus m_s can have two values: $+\frac{1}{2}$ (clockwise spinning) and $-\frac{1}{2}$ (anti-clockwise spinning). Arrow up ↑ is $+\frac{1}{2}$ (referred to as 'spin up') and arrow down ↓ is $-\frac{1}{2}$ (referred to as 'spin down').

Table of Allowed Quantum Numbers

n	l	m_l	Number of Orbitals	Orbital Name	Maximum Number of Electrons
1	0	0	1	$1s$	2
2	0	0	1	$2s$	2
	1	$-1,0,+1$	3	$2p$	6
3	0	0	1	$3s$	2
	1	$-1,0,+1$	3	$3p$	6
	2	$-2,-1,0,+1,+$	5	$3d$	10
4	0	0	1	$4s$	2
	1	$-1,0,+1$	3	$4p$	6
	2	$-2,-1,0,+1,+$	5	$4d$	10
	3	$-3,-2,-1,0,+$	7	$4f$	14

The Pauli Exclusion Principle

No two electrons in an atom can have identical quantum numbers. This means that two electrons occupying the same orbital can have maximum three quantum numbers (n, l, m_l)

identical, but the fourth quantum number, m_s, must be different such that two electrons in the same orbital must have opposite spins (↑↓).

The values of four quantum numbers for two electrons in a helium atom ($1s^2$) are shown below:

Electron	n	l	m_l	m_s
First electron	1	0	0	$+\frac{1}{2}$
Second electron	1	0	0	$-\frac{1}{2}$

Thus it is seen that the values for the first three quantum numbers are same but the fourth one is different.

Hund's Rule of Maximum Multiplicity

Electrons occupy the orbitals of a subshell (p, d, f subshells) in such a way as to give the maximum number of unpaired electrons and these unpaired electrons have the same direction of spins. This rule can be illustrated by considering the example of nitrogen.

Since electrons are charged particles and repel one another, the electron-electron repulsions are minimum when the electrons are as far apart as possible with parallel spins. Thus, state (a) has minimum repulsions and corresponds to lower energy (stable state). This is in accordance with Hund's rule. This principle is very important in guiding the filling of orbitals of $p, d,$ and f subshells, which have more than one type of orbital.

Electronic Configuration: Order of Filling the Orbitals

The Aufbau Principle: The orbitals are filled up in the order of increasing energy, i.e., the orbital of the lowest energy is filled up first while that of the highest energy is filled up in the end.

The order of filling of orbitals:

$$1s < 2s < 2p < 3s < 3p < 4s < 3d < 4p < 5s < 4d < 5p < 6s < 4f < 5d < 6p < 7s < 5f < 6d < 7p < 8s \ldots$$

$(n + l)$ **Rule:**

10. A new electron occupies the orbital for which $(n + l)$ is minimum. If we consider $3d$ and $4s$ orbitals, the electron will first enter the $4s$ orbital. This is because the value of $(n + l)$ for $4s$ is less $(4 + 0 = 4)$ than that for $3d$ orbital $(3 + 2 = 5)$.

11. If $(n + l)$ values for two orbitals are the same, the new electron enters the orbital for which n is minimum. In the case of $4f$ and $5d$ orbitals for which $(n + l)$ values are the same ($4 + 3 = 7$ and $5 + 2 = 7$), the electron will prefer to occupy the $4f$ orbital.

Solutions to Problems

Problems related to Atomic Structure

1. Calculate the radius of 1st, 2nd, and 3rd orbits of He^+ and Li^{2+} ions. Given that Bohr radius $a_H = 0.529 \times 10^{-10}$ m.

Formula:

$$r_n = \left(\frac{n^2}{Z}\right) a_H$$

- **For He^+ ion ($Z = 2$):**
  - $n = 1: r_1 = \frac{1^2}{2} \times 0.529 \times 10^{-10}\ \text{m} = 0.2645 \times 10^{-10}\ \text{m} = \mathbf{0.02645}$ nm
  - $n = 2: r_2 = \frac{2^2}{2} \times 0.529 \times 10^{-10}\ \text{m} = 1.058 \times 10^{-10}\ \text{m} = \mathbf{0.1058}$ nm
  - $n = 3: r_3 = \frac{3^2}{2} \times 0.529 \times 10^{-10}\ \text{m} = 2.3805 \times 10^{-10}\ \text{m} = \mathbf{0.23805}$ nm
- **For Li^{2+} ion ($Z = 3$):**
  - $n = 1: r_1 = \frac{1^2}{3} \times 0.529 \times 10^{-10}\ \text{m} = 0.1763 \times 10^{-10}\ \text{m} = \mathbf{0.01763}$ nm
  - $n = 2: r_2 = \frac{2^2}{3} \times 0.529 \times 10^{-10}\ \text{m} = 0.7053 \times 10^{-10}\ \text{m} = \mathbf{0.07053}$ nm
  - $n = 3: r_3 = \frac{3^2}{3} \times 0.529 \times 10^{-10}\ \text{m} = 1.587 \times 10^{-10}\ \text{m} = \mathbf{0.1587}$ nm

---

2. What are the energies of $n = 1$ and $n = 2$ states of H-atom in (a) J/atom and (b) kJ mol^{-1}? Given that $R_H = 1.097 \times 10^7\ m^{-1}$.

Formula:

$$E_n = -\frac{R_H hc}{n^2}$$

Where $h = 6.626 \times 10^{-34}$ J · s, $c = 3.0 \times 10^8$ m/s.

$$R_H hc = (1.097 \times 10^7) \times (6.626 \times 10^{-34}) \times (3.0 \times 10^8) \approx 2.179 \times 10^{-18}\ \text{J/atom}$$

- **(a) In** J/atom**:**
  - For $n = 1$:

$$E_1 = -\frac{2.179 \times 10^{-18}}{1^2} = \mathbf{-2.179 \times 10^{-18}}\ \text{J/atom}$$

  - For $n = 2$:

$$E_2 = -\frac{2.179 \times 10^{-18}}{2^2} = \mathbf{-5.448 \times 10^{-19}}\ \text{J/atom}$$

- **(b) In** kJ mol^{-1}**:**

  Multiply by Avogadro's number ($N_A = 6.022 \times 10^{23}\ \text{mol}^{-1}$) and divide by 1000:

  - For $n = 1$:

$$E_1 = -2.179 \times 10^{-18} \times 6.022 \times 10^{23} \times 10^{-3} = \mathbf{-1312.2}\ \text{kJ mol}^{-1}$$

  - For $n = 2$:

$$E_2 = -5.448 \times 10^{-19} \times 6.022 \times 10^{23} \times 10^{-3} = \mathbf{-328.1}\ \text{kJ mol}^{-1}$$

---

3. Calculate the frequencies in wave numbers per meter and the wavelengths in nm of light emitted from H-atom when the electron undergoes a transition ($R_H = 1.097 \times 10^7\ \text{m}^{-1}$):

Formula:

$$\bar{\nu} = R_H\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right), \quad \lambda = \frac{1}{\bar{\nu}}$$

- **(i) From** $n = 3$ **to** $n = 1$**:**

$$\bar{\nu} = 1.097 \times 10^7\left(\frac{1}{1^2} - \frac{1}{3^2}\right) = 1.097 \times 10^7 \times \frac{8}{9} = \mathbf{9.751 \times 10^6}\ \text{m}^{-1}$$

$$\lambda = \frac{1}{9.751 \times 10^6\ \text{m}^{-1}} = 1.0255 \times 10^{-7}\ \text{m} = \mathbf{102.55}\ \text{nm}$$

- **(ii) From** $n = 3$ **to** $n = 2$**:**

$$\bar{\nu} = 1.097 \times 10^7\left(\frac{1}{2^2} - \frac{1}{3^2}\right) = 1.097 \times 10^7 \times \frac{5}{36} = \mathbf{1.524 \times 10^6}\ \text{m}^{-1}$$

$$\lambda = \frac{1}{1.524 \times 10^6\ \text{m}^{-1}} = 6.563 \times 10^{-7}\ \text{m} = \mathbf{656.3}\ \text{nm}$$

- **(iii) From $n = 4$ to $n = 2$:**

$$\bar{\nu} = 1.097 \times 10^7 \left(\frac{1}{2^2} - \frac{1}{4^2}\right) = 1.097 \times 10^7 \times \frac{3}{16} = \mathbf{2.057 \times 10^6}\ \text{m}^{-1}$$

$$\lambda = \frac{1}{2.057 \times 10^6\ \text{m}^{-1}} = 4.861 \times 10^{-7}\ \text{m} = \mathbf{486.1}\ \text{nm}$$

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4. What is the difference between energy levels of sodium atom if emitted light has a wavelength of 589 nm?

Formula:

$$\Delta E = \frac{hc}{\lambda}$$

$$\Delta E = \frac{(6.626 \times 10^{-34}\ \text{J} \cdot \text{s}) \times (3.0 \times 10^8\ \text{m/s})}{589 \times 10^{-9}\ \text{m}} = \mathbf{3.375 \times 10^{-19}}\ \text{J}$$

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5. Calculate the frequencies of 1st and 2nd spectral line of Lyman series.

For Lyman series, $n_1 = 1$.

- **1st spectral line ($n_2 = 2$):**

$$\bar{\nu} = 1.097 \times 10^7 \left(1 - \frac{1}{4}\right) = \mathbf{8.228 \times 10^6}\ \text{m}^{-1}$$

$$\text{Frequency } (\nu) = c \cdot \bar{\nu} = (3 \times 10^8) \times (8.228 \times 10^6) = \mathbf{2.468 \times 10^{15}}\ \text{Hz}$$

- **2nd spectral line ($n_2 = 3$):**

$$\bar{\nu} = 1.097 \times 10^7 \left(1 - \frac{1}{9}\right) = \mathbf{9.751 \times 10^6}\ \text{m}^{-1}$$

$$\text{Frequency } (\nu) = c \cdot \bar{\nu} = (3 \times 10^8) \times (9.751 \times 10^6) = \mathbf{2.925 \times 10^{15}}\ \text{Hz}$$

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6. Calculate the frequencies of 1st and 2nd spectral line of Balmer series.

For Balmer series, $n_1 = 2$.

- **1st spectral line ($n_2 = 3$):**

$$\bar{\nu} = 1.097 \times 10^7 \left(\frac{1}{4} - \frac{1}{9}\right) = \mathbf{1.524 \times 10^6}\ \text{m}^{-1}$$

$$\text{Frequency } (\nu) = (3 \times 10^8) \times (1.524 \times 10^6) = \mathbf{4.571 \times 10^{14}} \text{ Hz}$$

- **2nd spectral line ($n_2 = 4$):**

$$\bar{\nu} = 1.097 \times 10^7 \left(\frac{1}{4} - \frac{1}{16}\right) = \mathbf{2.057 \times 10^6} \text{ m}^{-1}$$

$$\text{Frequency } (\nu) = (3 \times 10^8) \times (2.057 \times 10^6) = \mathbf{6.171 \times 10^{14}} \text{ Hz}$$

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7. Calculate the wavelength of an electron moving with a velocity of 2.5×10^7 ms^{-1}.

Formula:

$$\lambda = \frac{h}{mv}$$

Given: $m_e = 9.109 \times 10^{-31}$ kg, $v = 2.5 \times 10^7$ m/s, $h = 6.626 \times 10^{-34}$ J · s.

$$\lambda = \frac{6.626 \times 10^{-34}}{(9.109 \times 10^{-31}) \times (2.5 \times 10^7)} = \mathbf{2.91 \times 10^{-11}} \text{ m} = \mathbf{0.0291} \text{ nm}$$

Solutions to Conceptual Questions

12. **List the values of n, l, and m_l for orbitals in the 4d subshell.**
    - $n = 4$
    - $l = 2$
    - $m_l = -2, -1, 0, +1, +2$
13. **Give the values of the quantum numbers associated with the orbitals in the 3p subshell.**
    - $n = 3$
    - $l = 1$
    - $m_l = -1, 0, +1$
14. **What is the total number of orbitals associated with the principal quantum number $n = 3$?**
    - Total orbitals = $n^2 = 3^2 = \mathbf{9}$ (1 in 3s, 3 in 3p, 5 in 3d)
15. **What is the total number of orbitals associated with the principal quantum number $n = 4$?**
    - Total orbitals = $n^2 = 4^2 = \mathbf{16}$ (1 in 4s, 3 in 4p, 5 in 4d, 7 in 4f)

16. **What is the maximum number of electrons that can occupy the fourth energy level?**
    - Maximum electrons = $2n^2 = 2(4^2) = \mathbf{32}$
17. **When $n = 5$ what are the possible values for l?**
    - $l = 0,1,2,3,4$
18. **When $l = 2$ what are the possible values of m_l?**
    - $m_l = -2, -1, 0, +1, +2$
19. **How many sublevels are possible in the third energy level?**
    - **3** sublevels (3s, 3p, 3d)
20. **What is the maximum number of electrons that can occupy a d sublevel?**
    - **10** electrons $(2(2l + 1) = 2(2 \cdot 2 + 1) = 10)$
21. **What is the maximum number of electrons that can occupy a g sublevel?**
    - For g sublevel, $l = 4$.
    - Maximum electrons = $2(2(4) + 1) = \mathbf{18}$ electrons.
22. **When $n = 2$, the values of l can be** 0 **and** 1**.**
23. **When $l = 1$, the values of m_l can be** -1, 0, +1 **and the subshell has the letter designation of** p**.**
24. **When a subshell is labeled s, the value of l is** 0 **and m_l has value(s) of** 0**.**
25. **When a subshell is labeled p,** 3 **orbitals occur within the subshell.**
26. **When a subshell is labeled f, there are** 7 **values of m_l and** 7 **orbitals occur within the subshell.**